ClonoShift Resolves Robust Clonal Trajectories from Model-Dependent Uncertainty in Sparse Immune Repertoires

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Abstract

Longitudinal adaptive immune-repertoire sequencing enables quantitative tracking of clonal expansion, contraction, and persistence. However, sparse counts, changing library depths, and technical noise create severe inferential uncertainty, making conclusions sensitive to analytical choices: observation conditioning, response priors, and posterior summaries. To make this model dependence transparent rather than forcing an artificial consensus, ClonoShift evaluates the same qualified count input with two complementary models running by default: ClonoShift Extended, an observation-conditioned model developed in this work that evaluates symmetric midpoint coordinates and a symmetric Laplace response slab evaluated on the qualified response grid, and ClonoShift Core, a NoisET-compatible Rust implementation that preserves established reference conditioning and discrete response surfaces. Numerical validation confirms that Extended matches an independent 100-digit decimal oracle across all 972 evaluated count pairs spanning four classes (with exact 95% credible interval endpoints), while Core reproduces reference Python NoisET outputs within strict numerical tolerances (all maximum absolute differences below 2.0×10^{-10}). In matched benchmarks on an Apple M4 Pro system, Core completes response inference for 1,055,602 TCR clonotypes in a median 5.276 seconds (`18.6x` faster than pinned Python NoisET at 98.311 seconds), while the integrated dual-model product processes the million-clonotype repertoire end-to-end in a median 15.841 seconds (5.452 GB peak memory). Longitudinal TCR applications demonstrate concordance on large expansions and contractions while exposing summary sensitivity for sparse counts. In exact-pair B-cell receptor (BCR) tracking, empirical frequency scores discriminate recombinant hemagglutinin binding across lymph node contrasts, while calibrated response inference is properly refused because assay-matched technical calibration is absent. Crucially, technical noise and biological heterogeneity can generate mathematically identical negative binomial count distributions, establishing that sequencing counts alone cannot identify the causal source of overdispersion. ClonoShift reports model-dependent evidence of repertoire dynamics without claiming general biological false-discovery control. The software is licensed under GPL-3.0-or-later; repository access is private pending authorized public release.

1. Introduction

Longitudinal sequencing of adaptive immune repertoires connects sequencing counts to fundamental questions in immunology. By profiling T-cell receptor (TCR) and B-cell receptor (BCR) repertoires across successive timepoints, researchers can identify which clonotypes expand during acute infection, which contract into memory pools, and which antigen-specific lineages persist and hypermutate following vaccination [1,2].

However, high-throughput repertoire sequencing samples only a small fraction of circulating lymphocytes. Most clonotypes appear as sparse counts or remain undetected in individual libraries. Consequently, an observed count increase from zero to five can reflect true biological expansion, increased sequencing depth, or random Poisson sampling fluctuations. Distinguishing genuine biological changes from technical sampling noise remains the central statistical challenge of longitudinal repertoire analysis.

Probabilistic frameworks address this challenge by explicitly modeling latent clone frequencies alongside sampling variability. Pioneering work by Puelma Touzel and colleagues, implemented in the NoisET framework, established a Bayesian approach that couples power-law clone frequency distributions with Poisson observation models to infer repertoire change [1,2]. Beta-binomial formulations provide an alternative strategy for modeling overdispersion across replicate libraries [3].

Yet statistical inference under these frameworks is sensitive to analytical assumptions. Inferential conclusions can change depending on how unobserved counts are conditioned, whether response priors are symmetric around zero, and whether posterior distributions are summarized by means, medians, or directional tail probabilities. When alternative formulations yield differing conclusions for sparse clonotypes, forcing an artificial consensus or hiding model choice obscures essential statistical uncertainty. Investigators need to see that sensitivity directly rather than rely on opaque summary calls.

Furthermore, modern repertoire datasets frequently encompass hundreds of thousands or millions of unique clonotypes across longitudinal cohorts. This scale creates a strict operational requirement for fast, memory-efficient, deterministic, and reproducible computation. Reference implementations in interpreted languages can become computational bottlenecks when scaling to large studies.

ClonoShift addresses these statistical and computational challenges by evaluating the same qualified count input with two complementary probabilistic models. ClonoShift Extended implements an observation-conditioned model with a symmetric midpoint formulation and a symmetric Laplace response slab evaluated on the qualified response grid, importing qualified Core-derived noise parameters. ClonoShift Core provides a NoisET-compatible Rust implementation that preserves established reference conditioning and discrete response surfaces. Rather than selecting a single model or averaging discordant outputs, ClonoShift presents Extended and Core as balanced defaults. Agreement between the models supports robustness across the evaluated formulations, whereas disagreement highlights sensitivity to analytical assumptions or posterior summaries, directing the investigator to examine low-count uncertainty.

Here, we present the ClonoShift framework, its numerical validation, and its biological evaluation. We demonstrate that Extended matches an independent 100-digit decimal oracle and Core reproduces reference compatibility benchmarks across gDNA, TCR, and BCR datasets. We evaluate performance through strictly separated matched compatibility benchmarks and integrated dual-model product measurements on an Apple M4 Pro system. We evaluate ClonoShift on longitudinal TCR vaccination data and exact-pair recombinant BCR binding measurements, and we define the mathematical boundary of count-only inference, showing that technical and biological overdispersion can generate identical marginal count distributions.

2. Methods and implementation

2.1 Integrated ClonoShift architecture

ClonoShift executes both models from a single command: `clonoshift analyze --counts FILE --output NEW_DIRECTORY`. Ingestion validates the merged count table, preserves exact nucleotide CDR3 keys, and aligns counts deterministically. Both models receive the same validated count pairs, and their outputs are written to dedicated subdirectories before generating a side-by-side comparative summary (Figure 1):

```
results/
├── extended/
├── core/
└── comparison/
```

Count validation enforces positive library depths, valid integer counts, duplicate key rejection, and consistent amino-acid annotations. If one model encounters numerical refusal on an extreme observation, the valid result from the other model is preserved.

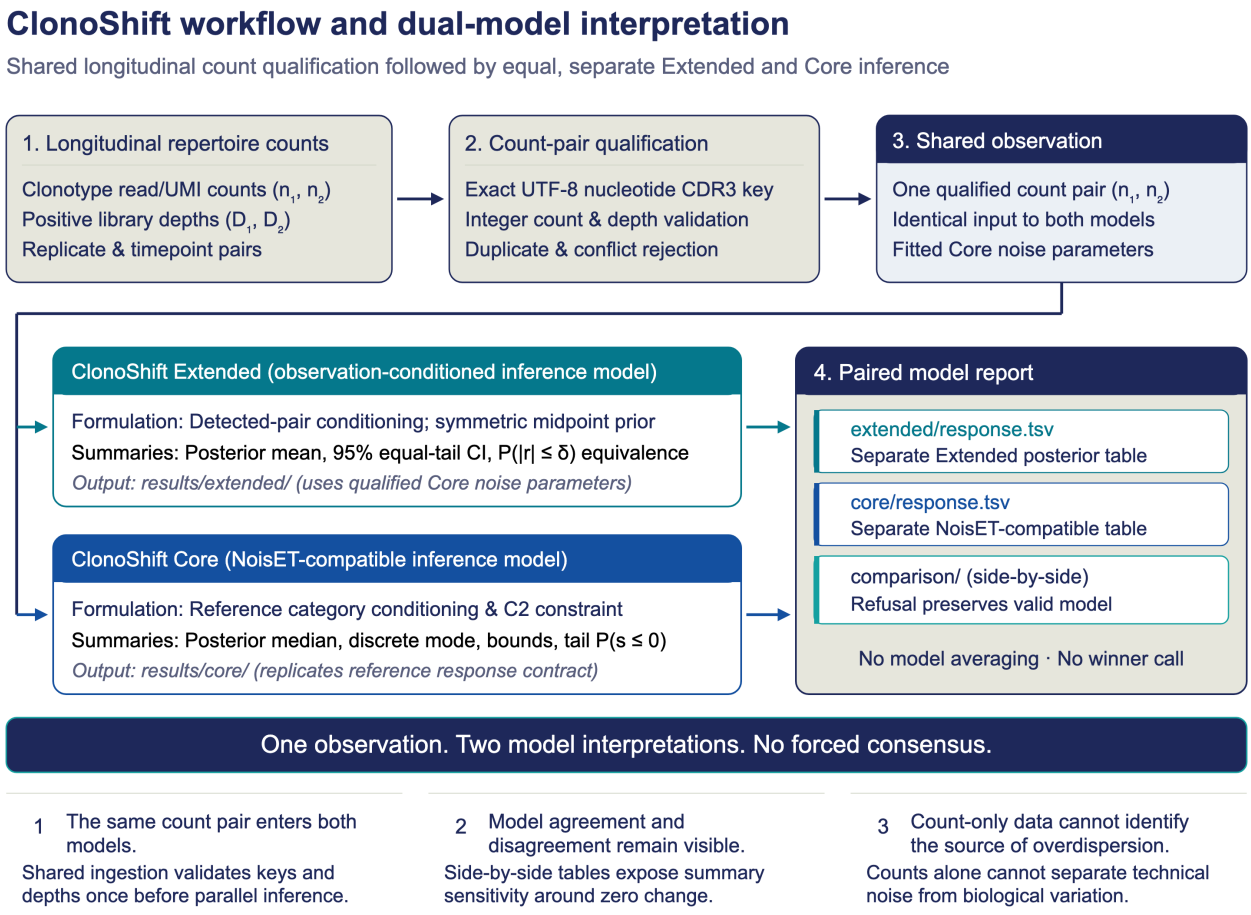


Figure 1. ClonoShift workflow and dual-model interpretation. Qualified counts and library depths feed the same observation to Extended (deep-teal header; observation-conditioned inference model) and Core (blue header; NoisET-compatible inference model) with equal presentation. Each model retains its own conditioning, prior, and posterior summaries (Methods 2.2–2.3). Separate outputs feed a descriptive paired report; a numerical refusal in one model preserves valid output from the other. Agreement and disagreement remain visible without model averaging or selecting a winner: *One observation. Two model interpretations. No forced consensus.* Count-only data cannot identify the source of overdispersion.

2.2 ClonoShift Extended

ClonoShift Extended is the observation-conditioned inference model developed in this work. It imports qualified Core-derived power-law parameters and evaluates a corrected Poisson observation model over a symmetric prior support.

To parameterize changes in latent clonotype frequencies f_1 and f_2 between two timepoints, Extended defines midpoint and log-response coordinates:

$$u = \log\sqrt{f_1 f_2}, \qquad r = \log(f_2 / f_1)$$

In this coordinate system, u is the log of the geometric mean frequency, while r represents the fold change on a natural logarithmic scale, such that $f_1 = e^{u - r/2}$ and $f_2 = e^{u + r/2}$.

Extended specifies a symmetric prior over (u, r) that combines an exact point mass at zero response with a symmetric Laplace response slab evaluated on the qualified response grid. On each response slice, the midpoint prior incorporates the frequency power-law density, coordinate Jacobian, and trapezoid integration weights, normalized over admitted support. The posterior state mass is proportional to:

$$\pi_{ur}\ \mathrm{Pois}(n_1 ; D_1\, e^{u - r/2})\mathrm{Pois}(n_2 ; D_2\, e^{u + r/2})$$

where n_1, n_2 are observed counts and D_1, D_2 are sequencing depth exposures. For detected pairs ($n_1 + n_2 > 0$), the integrated count likelihood is divided by the prior-predictive probability of observing a non-zero pair, which cancels in the posterior under fixed parameters. The reported per-clonotype `log_evidence` retains the unconditioned joint-prior normalizer; corrected null objectives and experimental response-prior objectives apply their corresponding observation normalizations separately.

Extended reports the posterior mean log-response, 95% equal-tail credible intervals, and the posterior probabilities of practical expansion ($r > \delta$), practical contraction ($r < -\delta$), and practical equivalence ($|r| \leq \delta$) under a declared practical effect threshold $\delta = 0.3$.

2.3 ClonoShift Core

ClonoShift Core is a NoisET-compatible Rust implementation of the reference Poisson noise and response framework [1,2]. It preserves the established first-observed-category conditioning, reported response bounds, and C2 constraint behaviors within an evaluated numerical scope.

Core fits frequency distribution parameters to replicate count pairs and computes discrete response surfaces. For each clonotype, Core reports the posterior median log-response, discrete mode, credible bounds, and the zero-inclusive nonpositive tail probability $P(s \leq 0)$. Crucially, because $P(s \leq 0)$ includes the zero-change atom, it describes nonpositive tail mass rather than a pure contraction probability, and Core does not compute Extended's practical-equivalence band probability $P(|r| \leq \delta)$.

2.4 Model comparison and failure semantics

ClonoShift model comparison writes side-by-side tables contrasting Extended and Core outputs without computing a forced consensus, a winner call, or a model-averaged estimate. The comparison categorizes directional agreement by evaluating the sign of Extended's posterior mean alongside Core's posterior median.

Extended reports a posterior mean over its qualified response grid, whereas Core reports a discrete posterior median; a small non-zero mean paired with a zero median is categorized as a sign difference. This distinction reflects summary sensitivity around zero rather than opposing directional evidence.

ClonoShift classifies every clonotype into explicit outcome states: `COMPLETED`, `REFUSED`, `FAILED`, and `UNAVAILABLE`. Shared input errors halt execution immediately. Numerical refusal by one model preserves the valid result from the other, allowing researchers to evaluate partial results transparently.

2.5 Scalable numerical implementation

ClonoShift is implemented in Rust to deliver memory efficiency, concurrency, and deterministic reproducibility. Unique count pairs are processed in parallel, and output ordering is preserved deterministically.

To accelerate high-depth inference while guaranteeing mathematical fidelity, Extended implements certified numerical omission. Negligible grid states are excluded only when an explicit certificate bounds omitted contributions within strict budgets (2^{-80} for mass, regions, relative evidence, and log evidence; 2^{-64} for the mean). Output publication uses atomic directory operations to prevent partial writes.

2.6 Experimental methods

Beyond the supported Poisson workflow, several experimental formulations were evaluated to investigate model behavior:

- **Negative Binomial (NB2) fitting:** Automatic corrected fitting and Negative Binomial (NB2) models with variance $\mu + \phi\mu^2$;
- **Parameter uncertainty propagation:** Joint bootstrap refitting of frequency and dispersion parameters across resampled clone pairs;
- **Abundance-aware allocation models:** Beta-binomial allocation models conditioned on total count $t = n_1 + n_2$;
- **Empirical-rank calibration:** Multiple-testing diagnostics evaluated using Benjamini–Hochberg and Benjamini–Yekutieli procedures [8,9].

These exploratory formulations were evaluated to assess model limits and are not part of the default production workflow.

2.7 Datasets and analytical units

ClonoShift was evaluated across four distinct biological and technical datasets:

- **Characterized gDNA technical replicates:** Four aliquots from a single blood-draw pool associated with Rytlewski et al. [3], evaluated as complete nucleotide fragments;
- **Longitudinal COVID-19 TCR tracking:** Donor M from Minervina et al. [4] across days 15, 37, and 85, evaluated using UMI-consensus CDR3β nucleotide keys;
- **Yellow fever vaccination TCR:** Donor S1 from Pogorelyy et al. [5] at days 0 and 15, linked to independent YFV and CMV antigen-reference records;
- **Paired heavy/light chain BCR dataset:** Single-cell dataset from Matz/Zhou et al. [7] encompassing 254,339 paired cells, 3,021 tested recombinant antibodies with measured hemagglutinin (HA) binding, and 14 longitudinal same-compartment contrasts (with earlier heavy-chain compatibility checked on Kim/Zhou et al. [6]).

Analytical units are strictly distinguished across all analyses: complete nucleotide sequences are distinct from amino-acid CDR3 clonotypes, and exact heavy/light BCR pairs are distinct from broader BCR clonal families. Assayed recombinant binding phenotypes apply strictly to tested exact pairs and are never propagated to unassayed clonal-family relatives.

2.8 Benchmark design

Performance benchmarking on an Apple M4 Pro system (12 logical CPUs, 24 GiB physical memory, `macOS27.0.1 build26A434`, `arm64`) strictly separates two benchmark scopes:

1. **Matched compatibility benchmark:** Compares single-model ClonoShift Core (`clonoshift legacy-response`) directly against the pinned Python NoisET implementation (`reference/upstream/NoisET` commit `2b6a45fcd26e7f9eefb3004d81725355cf6d66c9` via `scripts/noiset_matched.py --stage response`) using identical input tables, identical supplied noise parameters, identical response grid configuration, and identical output tables.
2. **Integrated product benchmark:** Measures the complete dual-model ClonoShift workflow (same-study replicate parameter fitting, configuration generation, parallel Extended and Core inference, and side-by-side comparison table publication). Because full dual-model ClonoShift performs replicate noise fitting and executes two distinct models, it is never compared directly as a speed ratio against single-model Python NoisET.

Each benchmark campaign executed one initial warmup run followed by three formal timed repetitions per tool and workload (`gdna`: 280,051 union clonotypes, 972 distinct count pairs; `million-real`: 1,055,602 union clonotypes, 3,614 distinct count pairs). Every started formal run was preserved without replacement. Timed execution measured end-to-end child process execution (`os.wait4`), and peak resident memory was recorded as the child process `wait4` peak resident set size (RSS, where $\mathrm{GB} = 10^9$ bytes). Independent scientific output validation and external cryptographic hashing occurred outside the timed interval. Speed ratios for the matched compatibility benchmark were calculated strictly from formal medians (Python NoisET median/ClonoShift Core median).

3. Results

3.1 Reference agreement, matched compatibility speed, integrated dual-model runtime, and model comparison

Both ClonoShift Extended and ClonoShift Core reproduce their respective numerical references within strict tolerances while maintaining a clear separation between numerical reference fidelity, matched single-model compatibility performance, integrated dual-model product runtime, and cross-model descriptive agreement (Figure 2; Tables S3 and S4).

First, ClonoShift Extended achieves high numerical agreement with an independent 100-digit decimal oracle across all 972 distinct count pairs (280,051 observations) in the characterized gDNA repertoire. Maximum absolute differences were 4.39×10^{-15} for the posterior mean, 1.40×10^{-14} for log evidence, 8.14×10^{-16} for $P(r > \delta)$, 1.39×10^{-15} for $P(r < -\delta)$, and 8.17×10^{-16} for $P(|r| \leq \delta)$. All discrete 95% credible interval endpoints matched exactly across expanding (468 pairs; 4,649 observations), contracting (440 pairs; 4,075 observations), zero-compatible (25 pairs; 262,201 observations), and equal nonzero (39 pairs; 9,126 observations) count-pair classes (Figure 2a; Figure S1).

Second, ClonoShift Core preserves numerical compatibility with the pinned Python NoisET reference across five evaluated datasets: gDNA replicates ($n = 972$ distinct pairs), longitudinal COVID-19 TCR day 15→37 ($n = 3,507$) and day 15→85 ($n = 3,614$), yellow fever vaccination TCR day 0→15 ($n = 1,910$), and heavy-chain BCR ($n = 53$). Across all datasets, maximum absolute discrepancies ranged from 3.55×10^{-15} to 1.43×10^{-13} for the posterior mean and from 1.22×10^{-15} to 1.42×10^{-12} for the zero-inclusive tail probability $P(s \leq 0)$ (Figure 2b; Table S4). Every dataset satisfied the frozen acceptance threshold of 2.0×10^{-10}, with exact matching of all discrete medians, modes, credible bounds, and selected key sets.

Third, in the matched compatibility benchmark comparing single-model Core against pinned Python NoisET on the Apple M4 Pro system (Figure 2c; Table S3), ClonoShift Core completed the 280,051-clonotype gDNA workload in a median 1.437 seconds (3-run range: 1.422–1.553 s; median peak RSS: 0.278 GB) compared with 28.986 seconds for Python NoisET (range: 28.899–31.466 s; median peak RSS: 0.929 GB), yielding a median speed ratio of `20.2x` (`20.171x`) and `3.34x` lower peak memory. On the 1,055,602-clonotype longitudinal TCR workload, Core completed inference in a median 5.276 seconds (range: 5.157–5.618 s; median peak RSS: 0.848 GB) compared with 98.311 seconds for Python NoisET (range: 97.935–99.277 s; median peak RSS: 2.727 GB), yielding a median speed ratio of `18.6x` (`18.634x`) and `3.22x` lower peak memory. Independent validation confirmed 100% scientific parity across all 280,051 and 1,055,602 rows, exact key ordering, zero nonfinite values, exact median and interval agreement on all count pairs, exact expanded-key selection (`2/2` and `4,249/4,249`), and byte-identical reproducibility across repetitions.

Fourth, in the integrated dual-model product benchmark measuring the full ClonoShift Extended + Core workflow (replicate noise fitting, Extended inference, Core inference, and comparison table publication; Figure 2d; Table S3), ClonoShift completed the 280,051-key gDNA workload in a median 4.607 seconds (range: 4.550–4.794 s; median peak RSS: 1.639 GB, range 1.639–1.651 GB) and the 1,055,602-key TCR repertoire in a median 15.841 seconds (range: 15.544–16.328 s; median peak RSS: 5.452 GB, range 5.130–5.545 GB). All formal repetitions produced byte-identical output tables. Because full dual-model ClonoShift executes noise parameter estimation alongside two distinct inference passes, this end-to-end product benchmark represents a broader scope than single-model response evaluation and is reported separately.

Finally, in the descriptive comparison between Extended and Core across all 1,055,602 observed clonotypes in the COVID-19 TCR repertoire (Figure 2e), ClonoShift recorded 686,246 sign agreements (65.0%) and 369,356 sign-category differences (35.0%), with zero unavailable rows. An exhaustive audit confirmed zero opposing non-zero signs (Extended mean $\times$ Core median $< 0 = 0$). Every difference paired a zero location summary in one model with a small non-zero summary in the other. Because Extended and Core use different observation conditioning, priors, and location summaries, these differences describe summary sensitivity in sparse-count regimes rather than model failure or opposing biological conclusions.

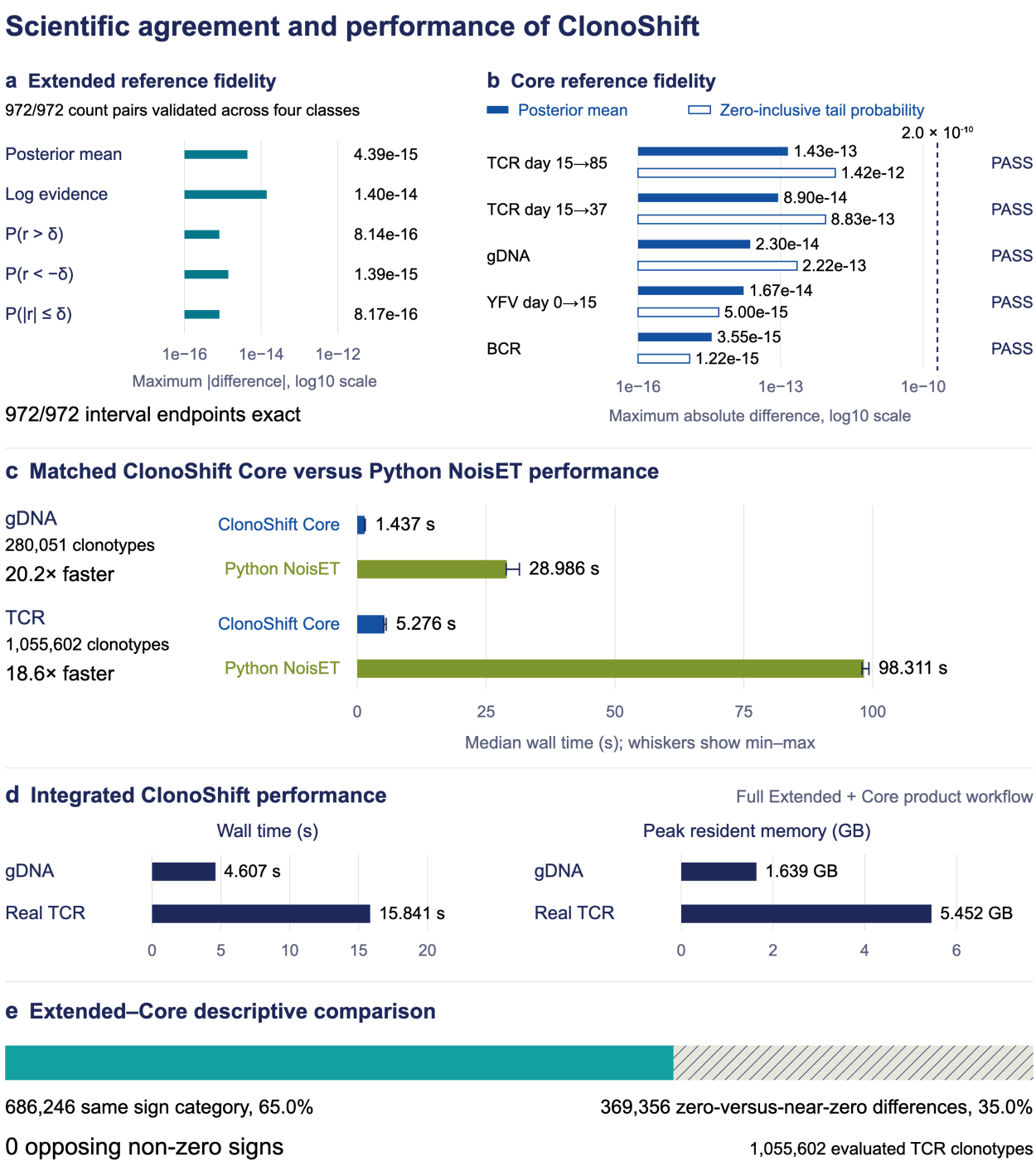


Figure 2. Scientific agreement and performance of ClonoShift. (a) Extended-versus-100-digit-oracle maximum absolute differences on a log10 scale across 972 gDNA count pairs, with exact credible interval endpoints (Figure S1). **(b)** Core-versus-pinned-Python-NoisET discrepancy bars for posterior mean (solid blue) and zero-inclusive tail probability $P(s \leq 0)$ (blue outline) across five datasets below the 2.0×10^{-10} threshold (dashed line). **(c)** Matched median wall time for Core (blue) versus Python NoisET (olive) on Apple M4 Pro under identical supplied noise parameters: `20.2x` faster for gDNA (1.437 s versus 28.986 s) and `18.6x` faster for 1,055,602 TCR clonotypes (5.276 s versus 98.311 s). **(d)** Integrated dual-model product workflow (replicate noise fitting, both inferences, comparison tables) running in 4.607 s (1.639 GB peak RSS) for gDNA and 15.841 s (5.452 GB peak RSS) for TCR. **(e)** Stacked bar of 1,055,602 TCR clonotypes showing 686,246 same-sign pairs (65.0%) and 369,356 zero-versus-near-zero differences (35.0%), with 0 opposing non-zero signs. Differences describe summary sensitivity, not model failure.

3.2 Representative count pairs show shared direction and summary sensitivity

ClonoShift Extended and ClonoShift Core agree on the direction of substantial count changes while providing distinct uncertainty summaries for sparse observations (Figure 3; Table S2).

In longitudinal TCR data from donor M (day 15 F1 to day 85 F1), high-magnitude count changes show concordant directionality between the two models. An expanding clonotype (`2 → 13` counts; key `TGCAAACACCGACTAGCGGGAAACACCGGGGAGCTGTTTTTT`) yielded an Extended posterior mean log-response of $+2.116$ with a 95% equal-tail credible interval of $[+0.944, +2.833]$ and practical-equivalence probability $P(|r| \leq 0.3) = 0.0244$ ($P(r > 0.3) = 0.976$), alongside a Core posterior median of $+1.512$ with reported bounds $[+0.302, +2.923]$ and nonpositive tail probability $P(s \leq 0) = 0.0103$ (Figure 3, expansion group). Conversely, a contracting clonotype (`44 → 3` counts; key `TGCAGCACGAACGGGGGGCTGGATCAGCCCCAGCATTTT`) yielded an Extended posterior mean of -2.899 with a 95% credible interval of $[-3.777, -1.889]$ and $P(|r| \leq 0.3) = 6.96 \times 10^{-9}$, alongside a Core posterior median of -2.016 with reported bounds $[-3.024, -1.109]$ and nonpositive tail probability $P(s \leq 0) = 1.000$ (Figure 3, contraction group). In both cases, the credible intervals from both models lie strictly on the same side of zero, supporting directional robustness.

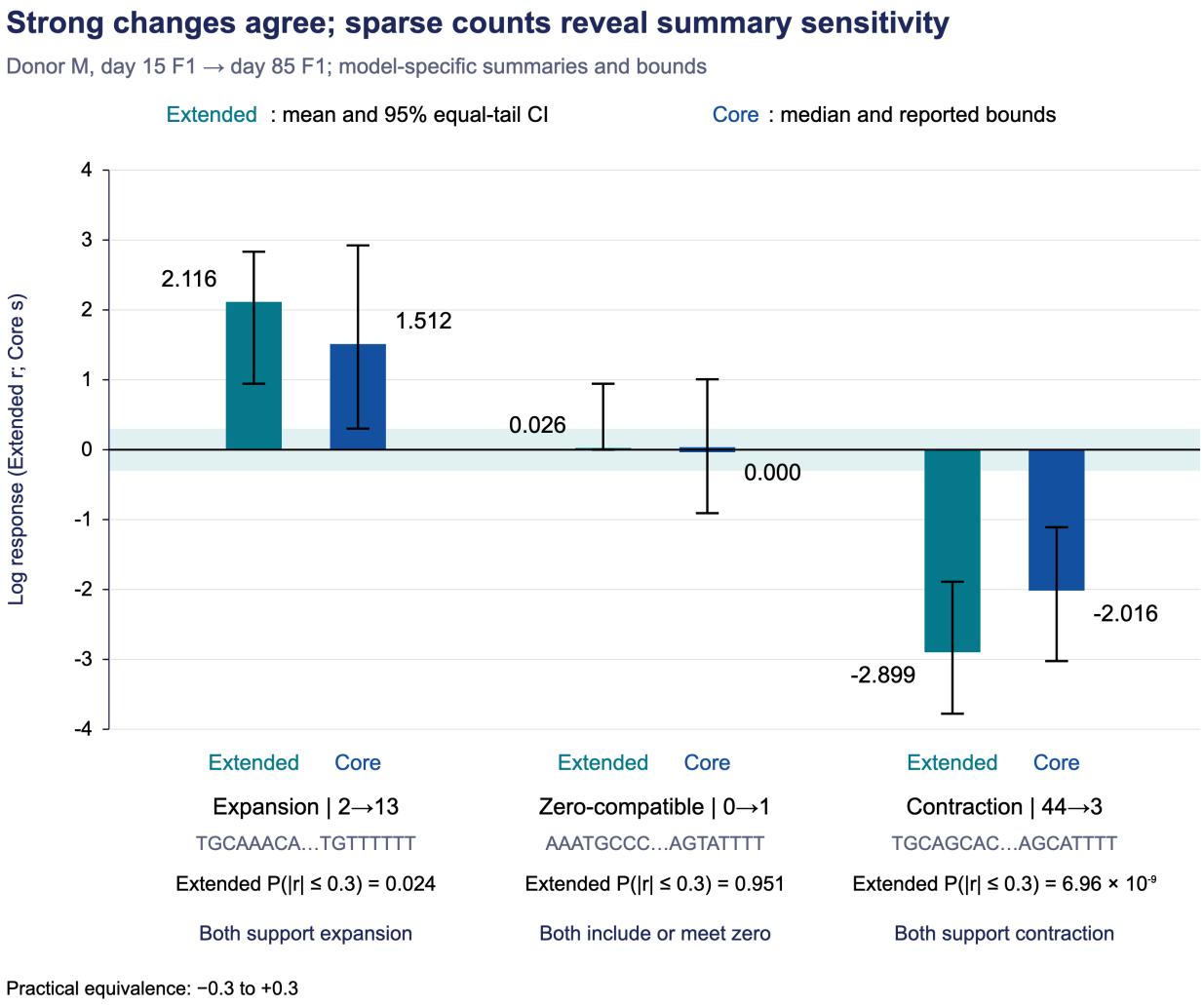


Figure 3. Strong changes agree; sparse counts reveal summary sensitivity. Grouped diverging vertical bars compare three representative nucleotide TCR examples from donor M (day 15 F1→day 85 F1): expansion (2→13), zero-compatible change (0→1), and contraction (44→3). Bar height is the Extended posterior mean (deep teal) or Core posterior median (blue), starting at zero. Capped vertical whiskers show the Extended 95% equal-tail credible interval or Core reported bounds. Black numerical labels indicate location summaries; the horizontal turquoise practical-equivalence band spans −0.3 to +0.3. Practical-equivalence probability is defined only for Extended. These are model-specific summaries and their direct comparison is descriptive; agreement is not independent biological validation. Full-precision values and sequences remain in Table S2.

Sparse observations highlight the distinct uncertainty representations of the two models. For a zero-compatible `0 → 1` count transition (key `AAATGCCCAGTGCTCCTCCTTTACACAGATACGCAGTATTTT`), Extended yielded a posterior mean log-response of +0.026 with a 95% credible interval of [0.000, +0.944] and assigned $P(|r| \leq 0.3) = 0.951$ to the practical-equivalence band (Figure 3, zero-compatible group). On the same count pair, Core yielded a discrete posterior median of 0.000 with reported bounds [−0.907, +1.008] and nonpositive tail mass $P(s \leq 0) = 0.604$. Similarly, for a `1 → 0` transition (key `AAAAAACTGTTTTTT`; Table S2), Extended yielded a posterior mean of −0.021 (95% CI: [−0.944, 0.000]) with $P(|r| \leq 0.3) = 0.953$, whereas Core yielded a median of −0.101 (bounds: [−1.512, +0.706]).

These sparse-count results reflect differences between model-specific location summaries rather than opposing biological directions. Core's median is zero because substantial posterior mass falls on the point mass at zero response. Extended reports a posterior mean over its qualified response grid, whereas Core reports a discrete posterior median. Extended computes its posterior mean from a symmetric Laplace response slab evaluated on the qualified response grid, producing a small non-zero value while

assigning high posterior probability to practical equivalence. Practical-equivalence probability is defined and reported only by Extended; Core characterizes low-count uncertainty through wide bounds spanning zero and its zero-inclusive tail probability $P(s \leq 0)$.

3.3 Real TCR analyses show concordant model outputs within clear boundaries

Application to real longitudinal TCR datasets demonstrates that ClonoShift captures clonal dynamics in vaccine and infection cohorts while maintaining explicit boundaries on biological interpretation (Figure 4; Table S7).

In yellow fever vaccination (donor S1, day 0 to day 15; 1,258,573 exact nucleotide keys) [5], ClonoShift evaluated clonotypes linked to independent antigen-reference records. Among 40 observed YFV-associated reference identities, Extended identified 5 (12.5%) as practical-effect expansions ($P(r > 0.3) > 0.95$ at 129 quadrature nodes), and Core identified the exact same 5 reference identities (12.5%) using its zero-effect positive-tail diagnostic ($P(s \leq 0) < 0.025$; Figure 4a). Among 12 observed CMV-associated specificity controls, both Extended and Core identified zero expansion hits (0/12, 0.0%).

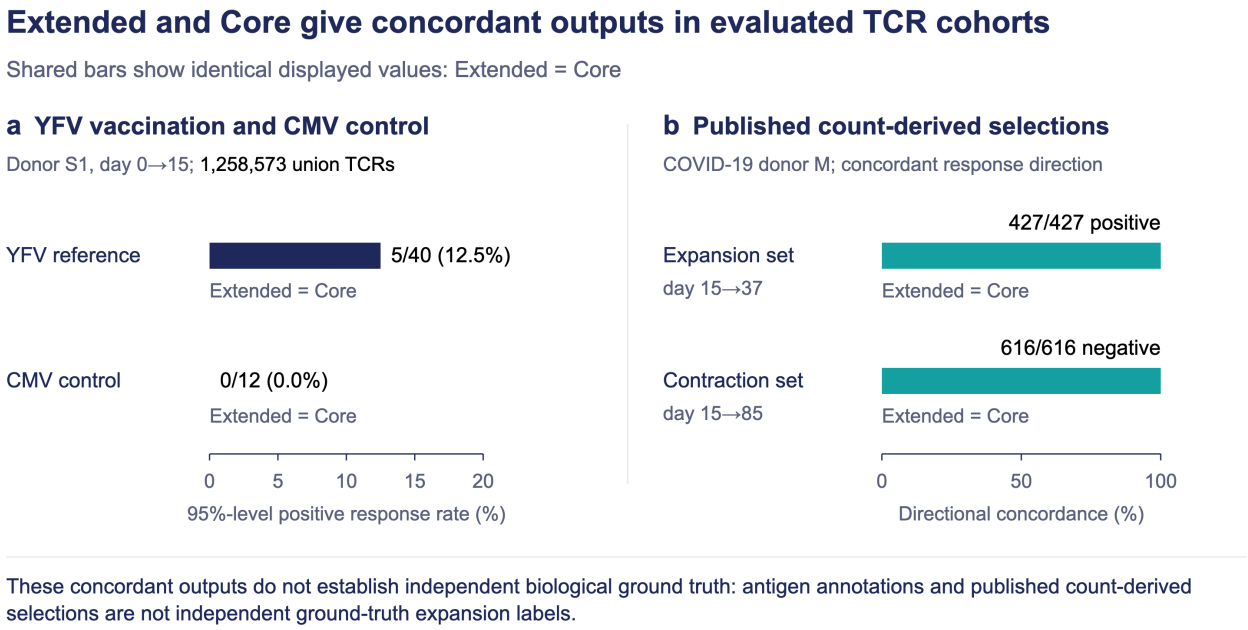


Figure 4. Extended and Core give concordant outputs in evaluated TCR cohorts. Shared bars show identical displayed outputs labeled "Extended = Core", while decision rules remain model-specific. **(a)** YFV-associated references and CMV controls in donor S1 [5]: Extended uses $P(r > 0.3) > 0.95$ at 129 quadrature nodes; Core uses $P(s \leq 0) < 0.025$. Both identify 5/40 YFV hits and 0/12 CMV hits. **(b)** Published count-derived COVID-19 selections in donor M [4]: all 427 expansion-associated clonotypes are positive and all 616 contraction-associated clonotypes negative in both models. These annotations and selections are not independent biological ground truth.

In longitudinal COVID-19 TCR tracking (donor M) [4], all 427 published expansion-associated clonotypes between days 15 and 37 exhibited positive posterior mean directions in both Extended (427/427, 100%) and Core (427/427, 100%), and all 616 published contraction-associated clonotypes between days 15 and 85 exhibited negative posterior mean directions in both Extended (616/616, 100%) and Core (616/616, 100%) (Figure 4b).

While these evaluations demonstrate high analytical concordance between Extended and Core, the evaluated annotations and published selections do not constitute exhaustive biological ground truth. External antigen-reference annotation establishes receptor specificity rather than guaranteed cellular

expansion in a given sampling window; uncalled YFV-annotated clonotypes may remain quiescent or below sequencing depth. Furthermore, published COVID-19 clonotype sets were originally defined by the intersection of count-based callers (`edgeR` ∩ `NoisET`) [4]. Concordance on these sets demonstrates analytical continuity across implementations rather than orthogonal biological validation.

3.4 Exact-pair BCR analyses illustrate compartment-dependent binding associations and need for assay-matched calibration

Evaluation of exact paired heavy/light B-cell receptors following quadrivalent influenza mRNA vaccination (254,339 paired cells; 3,021 assayed recombinant monoclonal antibodies with 1,070 HA-positive and 1,951 HA-negative ELISA phenotypes; 563 unique exact pairs across 573 occurrences in 14 same-compartment contrasts) [7] shows that empirical frequency changes track hemagglutinin (HA) binding in a donor- and compartment-dependent manner (Figure 5; Table S5).

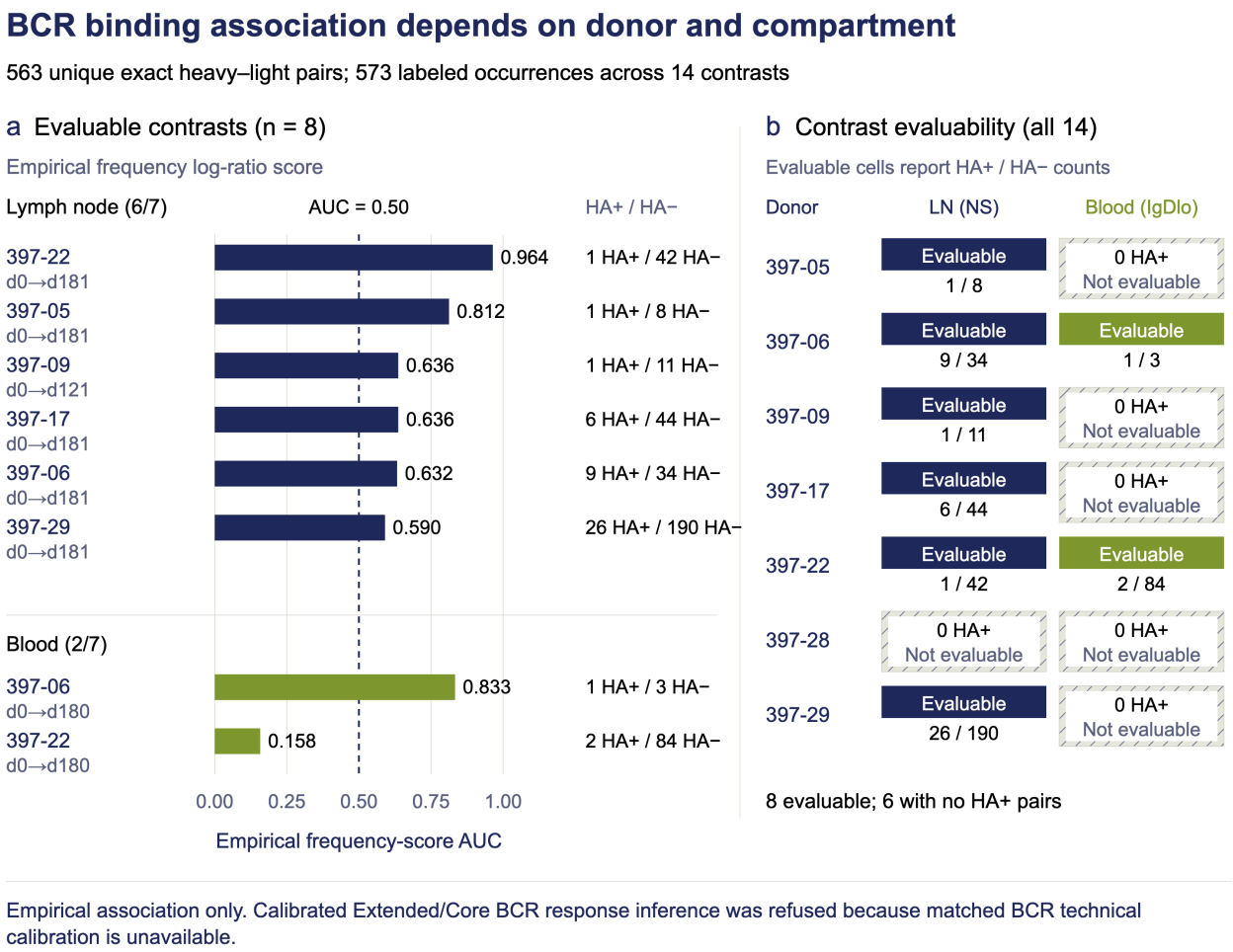


Figure 5. BCR binding association depends on donor and compartment. Empirical observed-frequency log-ratio scores (pseudocount 0.5) discriminate recombinant HA binding in tested exact heavy/light BCR pairs [7]; full contrast identifiers and counts are in Table S5. **(a)** Eight evaluable AUC bars in lymph node (navy, descending AUC) and blood (olive), with HA+/HA− counts and dashed AUC = 0.50 reference. **(b)** Evaluability matrix for all 14 donor-by-compartment contrasts (hatched cells indicate zero HA-positive pairs). Calibrated Extended/Core BCR response inference was refused because matched technical calibration is unavailable; binding labels are never propagated across clonal families.

Across the eight evaluable contrasts containing at least one assayed HA-positive antibody (Figure 5a), an empirical observed-frequency log-ratio score (with a 0.5 pseudocount) achieved AUC values above 0.50 in all six evaluable lymph node (`LN/NS`) contrasts (ranging from 0.590 in donor `397-29 (d0→d181)` to

0.964 in donor `397-22 (d0→d181)`). In contrast, only two of seven peripheral blood (`blood/IgDlo`) contrasts were evaluable, showing mixed discrimination: AUC = 0.833 in `397-06 (d0→d180)` with 1 HA+ and 3 HA− pairs versus AUC = 0.158 in `397-22 (d0→d180)` with 2 HA+ and 84 HA− pairs. Across all 14 same-compartment contrasts (7 donors × 2 compartments; Figure 5b), six contrasts (1 lymph node and 5 blood) contained zero qualifying HA-positive pairs (`0 HA+`) and were non-evaluable for AUC.

This analysis relies strictly on an empirical observed-frequency score: calibrated Extended and Core BCR response inference was formally refused because matched technical calibration was unavailable (qualification gate `REFUSED_FOR_CALIBRATED_BCR_RESPONSE`, Table S8). Specifically, the single-cell BCR dataset lacks technical replicate libraries (`tech_rep = 1` across all 109 sample records), gDNA noise parameters cannot be transferred across distinct sequencing technologies, and recombinant binding assays measure static molecular affinity rather than longitudinal clonal expansion. Furthermore, binding phenotypes apply strictly to assayed exact heavy/light chain pairs and are never propagated to unassayed members of the same clonal family.

3.5 Conceptual and experimental limits of count-only overdispersion inference

Theoretical and experimental analyses establish why sequencing counts alone cannot determine the causal origin of excess variability (Figure 6; Table S6).

When a gamma-distributed multiplier $G \sim \text{Gamma}(1/\phi, \phi)$ scales a Poisson sampling rate, the resulting marginal count distribution follows a Negative Binomial (`NB2`) law with variance $\text{Var}(K) = \mu + \phi\mu^2$ (Figure 6a,b). Mathematically, this identical marginal distribution emerges whether G originates from technical overdispersion (such as library preparation, PCR amplification, or capture efficiency; Figure 6a, top) or true biological heterogeneity across cellular draws (Figure 6a, bottom). Because both mechanisms generate the exact same count law, count data alone cannot identify the causal source of overdispersion.

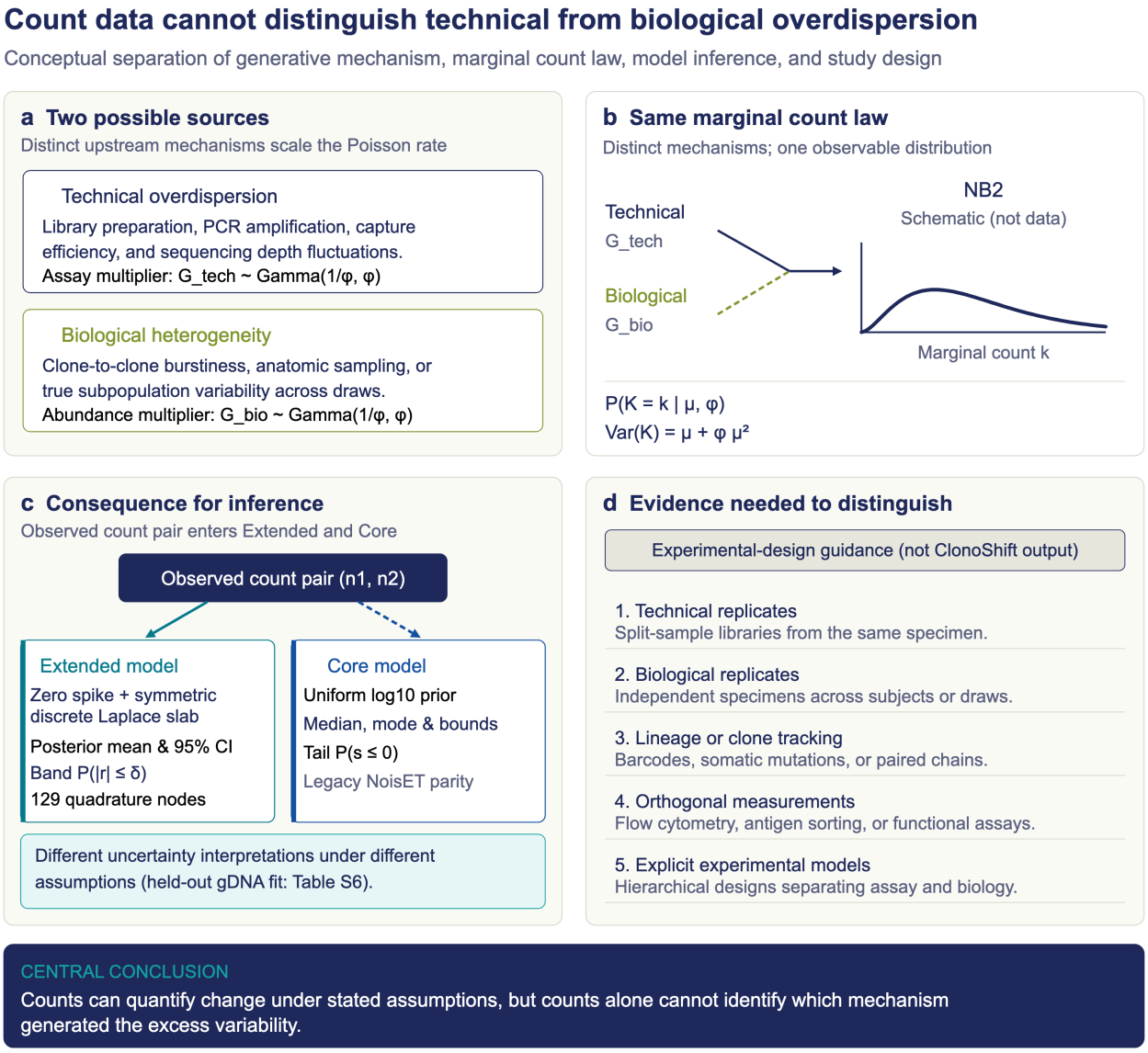


Figure 6. Count data cannot distinguish technical from biological overdispersion. (a) Technical and biological gamma-distributed rate multipliers represent distinct generative mechanisms. **(b)** Both converge on the same marginal NB2 count law; the curve is schematic, not empirical data. **(c)** The shared count pair feeds Extended (zero spike + symmetric discrete Laplace slab, posterior mean, 95% CI, and practical-equivalence probability) and Core (uniform log10 prior, median, mode, bounds, and tail $P(s \leq 0)$); held-out evaluations are in Table S6. **(d)** Technical and biological replicates, lineage tracking, and orthogonal measurements are study-design guidance, not ClonoShift outputs. Counts quantify change under stated assumptions but cannot identify which mechanism generated the excess variability.

When an observed count pair (n_1, n_2) is evaluated by ClonoShift Extended and ClonoShift Core, both models quantify uncertainty under their respective stated assumptions without forcing an artificial consensus (Figure 6c). Extended evaluates an exact point mass at zero response plus a symmetric Laplace response slab evaluated on the qualified response grid to report the posterior mean, 95% credible interval, and practical-equivalence probability $P(|r| \leq \delta)$, whereas Core evaluates a uniform $\log_{10}$ response prior to report the posterior median, mode, bounds, and nonpositive tail probability $P(s \leq 0)$.

In exploratory evaluations across six correlated held-out folds from four aliquots of the characterized gDNA pool (Table S6), abundance-binned and smooth beta-binomial allocation models reduced conditional held-out negative log-likelihood relative to a global baseline across every fold. However,

both models retained nonuniform two-sided tail rejection rates (1.27% to 12.58% at the nominal 5% level) among high-abundance observations ($t \geq 100$). Thus, improved statistical fit does not resolve the physical source of overdispersion or establish calibrated biological discovery. Distinguishing technical noise from biological dynamics requires explicit experimental designs: technical replicates, biological replicates, cellular lineage tracking, and orthogonal functional assays (Figure 6d). These represent experimental-design principles rather than outputs generated by ClonoShift software.

4. Discussion

ClonoShift introduces an integrated framework for longitudinal immune-repertoire analysis that applies two complementary probabilistic models to the same qualified count input. By evaluating ClonoShift Extended and ClonoShift Core side-by-side with balanced presentation, ClonoShift allows backward compatibility and an alternative methodological formulation to be evaluated together rather than forcing an artificial choice between them.

Paired reporting clarifies the interpretation of repertoire dynamics. When Extended and Core agree, that concordance supports robustness across the evaluated formulations. When the models differ, that divergence makes model dependence visible. In sparse-count regimes, differences between Extended and Core stem primarily from contrasting location summaries near zero: Extended reports a posterior mean over its qualified response grid, whereas Core reports a discrete posterior median (369,356 zero-versus-near-zero differences and 0 opposing non-zero signs across 1,055,602 TCR clonotypes). Rather than forcing a consensus call that would obscure this sensitivity, ClonoShift directs investigators to recognize when low-count data do not constrain directional inference.

The mathematical equivalence between Poisson–gamma compounding and the negative binomial distribution highlights a fundamental boundary: sequencing counts alone cannot reveal whether observed overdispersion reflects technical variation or biological dynamics. Improved goodness-of-fit does not identify the causal origin of excess noise. Consequently, technical replicates, biological replicates, and orthogonal measurements remain essential to disentangle experimental artifacts from immune responses.

ClonoShift v1.0.0 provides reproducible inference while maintaining explicit scientific boundaries (Table S8). The software does not claim universal biological expansion calls, empirical biological false-discovery rate (FDR) control, assay-independent calibration, or validated BCR clonal-family response inference. Where adequate calibration data are absent, such as in single-cell BCR datasets lacking technical replicates, ClonoShift correctly refuses calibrated inference. These boundaries reflect both the intrinsic limits of count-only repertoire sequencing and the calibration evidence available for the present study.

Implemented in Rust, ClonoShift delivers high computational performance and strict reproducibility. ClonoShift Core accelerates matched single-model response inference by `20.2x` on gDNA (median 1.437 s) and `18.6x` on real TCR repertoires (median 5.276 s) compared with reference Python NoisET on an Apple M4 Pro system. Furthermore, the complete integrated product workflow processes over one million clonotypes end-to-end in a median 15.841 seconds with 5.452 GB peak memory. Schema validation, certified numerical omission, atomic directory operations, and deterministic sorting ensure byte-identical outputs across repeated runs.

Future work should prioritize four areas: assay-matched technical replicate series for platform noise baselines; prospective validation cohorts with known cellular expansion; phylogenetic models of B-cell clonal-family structures; and empirical decision-calibration frameworks for translational applications.

5. Data and code availability

ClonoShift v1.0.0 is qualified under GPL-3.0-or-later as a source-only, data-free software package with synthetic test fixtures. Core is a NoisET-compatible Rust implementation and Extended is the observation-conditioned inference model developed in this work [1,2]. Benchmarking was conducted on an Apple M4 Pro system (`arm64`, `macOS27.0.1 build26A434`). The source code is maintained in the ClonoShift source repository; repository access remains private pending authorized public release. Archive DOI: none assigned (not available).

Biological datasets analysed in this study were obtained from original public repositories (Table S1): Minervina COVID-19 TCR (Zenodo 4065547, PRJNA633317) [4]; Pogorelyy yellow fever TCR (PRJNA493983) [5]; Kim/Zhou BCR (Zenodo 5895181, PRJNA777934) [6]; Matz/Zhou paired BCR (Zenodo 13929847, PRJNA1049392) [7]; and Rytlewski gDNA (PLoS ONE 14:e0213684) [3]. No restricted or proprietary biological data are redistributed in the software repository.

Author declarations

Author contributions: Yoon Sing Yap conceived the study, developed the software and methodology, performed the analyses and validation, interpreted the results, prepared the figures, and wrote and revised the manuscript.

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Competing interests: The author declares no competing interests.

Ethics and consent: This study performed secondary computational analyses of previously published, publicly available datasets. No new human participants were recruited, and no new biological specimens were collected. Ethics approvals and informed-consent procedures for the original studies are described in the corresponding source publications. This study did not attempt to re-identify participants.

References

1. Bensouda Koraichi M, Puelma Touzel M, Mazzolini A, Mora T, Walczak AM. NoisET: Noise learning and Expansion detection of T-cell receptors. arXiv:2102.03568 (2021; revised 2022). doi:10.48550/arXiv.2102.03568.
2. Puelma Touzel M, Walczak AM, Mora T. Inferring the immune response from repertoire sequencing. PLoS Computational Biology 16:e1007873 (2020). doi:10.1371/journal.pcbi.1007873.
3. Rytlewski J, Deng S, Xie T, Davis C, Robins H, Yusko E, Bienkowska J. Model to improve specificity for identification of clinically-relevant expanded T cells in peripheral blood. PLoS ONE 14:e0213684 (2019). doi:10.1371/journal.pone.0213684.
4. Minervina AA et al. Longitudinal high-throughput TCR repertoire profiling reveals the dynamics of T-cell memory formation after mild COVID-19 infection. eLife 10:e63502 (2021). doi:10.7554/eLife.63502.
5. Pogorelyy MV et al. Precise tracking of vaccine-responding T cell clones reveals convergent and personalized response in identical twins. PNAS (2018). doi:10.1073/pnas.1809642115.
6. Kim W, Zhou JQ, Horvath SC, et al. Germinal centre-driven maturation of B cell response to mRNA vaccination. Nature 604:141–145 (2022). doi:10.1038/s41586-022-04527-1.
7. Zhou JQ. mRNA-based influenza vaccine expands the breadth of the B cell response in humans. Processed data accompanying Matz et al., Nature Immunology (2026). Zenodo. doi:10.5281/zenodo.13929847.
8. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. Journal of the Royal Statistical Society B 57:289–300 (1995). doi:10.1111/j.2517-6161.1995.tb02031.x.
9. Benjamini Y, Yekutieli D. The control of the false discovery rate in multiple testing under dependency. Annals of Statistics 29:1165–1188 (2001). doi:10.1214/aos/1013699998.

Supplementary methods and tables

S1. Observed-pair inference and numerical interpretation

Let Q(n1,n2) be the joint count probability after summing the supplied joint prior over frequency/response states. The detected-pair law is Q(n1,n2)/(1−Q(0,0)) for n1+n2>0. The normalization is integrated over the prior; dividing every latent-state likelihood by its own detection probability would define a different posterior. At a fixed prior, global detection normalization cancels in the posterior over states. Extended's emitted per-clonotype log_evidence is log Q(n1,n2), rather than log of the detected-pair law. Observation-conditioned null objectives and the experimental response-prior objective have their respective integrated observation normalizations.

The finite response support is symmetric and constrained by the frequency limits. Midpoint-slice masses include the frequency density, the log-coordinate Jacobian and trapezoid weights. The response slab is proportional to $\exp(-|r|/b)$, including zero; adding the zero spike therefore makes the nonzero response mass differ from the supplied mixture weight. Equal-tail intervals invert the discrete CDF. Region events are strictly $r > \delta$ and $r < -\delta$, with the closed practical band between them.

The independent 100-digit oracle evaluated 972 distinct pairs representing 280,051 observations over 5,121 prior states. Maximum absolute log-evidence and mean differences were 1.4023934759953227e−14 and 4.390784319591717e−15, respectively. Every discrete interval endpoint matched. Omission budgets bound excluded-state mass/region/evidence contributions separately from floating-point error: mass, regions, relative evidence and log evidence use 2^{-80}, while the mean uses 2^{-64}. These are evaluated numerical contracts, not biological adequacy criteria.

S2. Compatibility and experimental uncertainty

Core numerical checks use the pinned NoisET formulation with supplied common noise parameters. Reference discrete response grids and selected outputs are compared separately from optimizer endpoints. Edge-case tests preserve known upstream behavior where required by compatibility. The historical edge-case record contains 63,403 numerical and 207 exact comparisons; its maximum fixed-evaluation discrepancy is 2.8446711652918566e−10. The primary fitted-objective difference is separately 2.0715175352847837e−7. Relative tolerance can admit a fixed-evaluation absolute discrepancy larger than the absolute tolerance; the applicable limits are not a universal absolute-error theorem.

Experimental parameter uncertainty resamples observed clone-pairs and jointly refits dispersion and frequency nuisance parameters. Each normalized response posterior has equal weight in the mixture. Target counts do not reweight the fitted training laws. The corrected NB2 response ensemble uses 12 observed-pair draws. Separate conditional-prediction sensitivity studies use 24 draws for bins and 12 for the smooth evaluator; these are different procedures. The mixture mean and total variance retain within-fit and between-fit components, and mixture quantiles are not averages of interval endpoints. Failure of a required refit remains refusal; a point estimate does not silently replace the requested ensemble. Unseen-population, true-exposure and clone-dependence uncertainty are outside this observed-pair procedure. Adaptive likelihood grids did not resolve all continuous tails or modes and do not supply a resolved parameter posterior.

S3. Identifiability and technical-null design

For $G_j \sim \text{Gamma}(1/\phi, \text{scale} = \phi)$ and $K_j \mid G_j \sim \text{Poisson}(\mu_j G_j)$, the marginal distribution is NB2. Its probability is $\frac{\Gamma(k + 1/\phi)}{\Gamma(1/\phi)\, k!} \frac{(\mu\phi)^k}{(1 + \mu\phi)^{k + 1/\phi}}$. The causal description of G_j does not appear in that law. Independently varying biological abundance and independently varying technical acquisition can thus produce observationally identical counts. This is an explicit counterexample to count-only causal identification, rather than a claim that all real variability follows this construction.

Under equal true exposures and unchanged latent frequency, conditioning independent NB2 counts on their total gives BetaBinomial(t, $1/\phi$, $1/\phi$). Totals zero and one provide no allocation-dispersion information. Equal observed library sums do not prove equal true exposure. The unequal-proxy conditional model uses $\alpha = 2p/\phi$ and $\beta = 2(1 - p)/\phi$ as its own allocation law. The abundance-bin model uses total-count bins [0,4), [4,10), [10,30), [30,100) and [100,1000001), with global fallback when fewer than 30 informative observations have total at least two in a stratum; the smooth model uses $\phi(t) = \beta t^{-\eta}$, $\eta \in [0, 2]$. Neither is assumed to define an assay-independent generative count model.

Four aliquots of one gDNA pool produce three disjoint pair matchings evaluated in both training/target orientations. The six folds reuse libraries and are not independent experimental replicates. The same small collection also informed model exploration. Library-total fractions proxy exposure. Conditional NLL evaluates held-out allocation prediction. Inclusive two-sided tails equal $\min\{1, 2\min[P(X \le x), P(X \ge x)]\}$; their empirical fractions are model-tail diagnostics. Overall rare-clone dominated fractions cannot replace abundance-stratified checks. The physical-null experiment does not include variation preceding the material split or establish transport to RNA or single-cell assays.

Experimental empirical-rank calibration uses disjoint fit/calibration/test identities assigned by SHA-256 key modulo ten: the first four digest bytes are interpreted as an unsigned big-endian integer before the modulo operation. Residues 0–4 fit, 5–7 calibrate and 8–9 test. Calibration and testing require total≥5. Within prespecified total-count strata, upper ranks equal (1 + number of calibration scores at least the target score)/(1 + number of calibration scores), retaining ties conservatively. Missing strata refuse. Exchangeability within strata is necessary and is not established generally across source distributions or assays. BH/BY adjustments preserve literal clone multiplicity; neither repairs invalid marginal null values. Posthoc practical-effect filters remain diagnostic.

S4. Biological annotation and exact receptor identity

Minervina comparisons retain literal nucleotide CDR3 identities and UMI-consensus count units. The published directional lists were count-derived intersections of edgeR and NoisET results. Their concordance is not an independent sensitivity estimate. For YFV, nucleotide identities remain the count unit, while exact CDR3beta amino acid plus compatible highest-score V and J calls determine the annotation link. One reference identity may match several nucleotide observations; an observed reference hit is not a pooled posterior over those observations. The original recovered reference contains 265 YFV and 371 CMV records, differing from narrative counts in the publication. Every actual record is retained. At confidence at least one, 37 YFV and 4 CMV reference identities are observed; at confidence at least three, 34 YFV and no CMV identities are observed. No high-confidence CMV comparison is invented.

In the exact-pair BCR study, heavy and light rows link by source cell_id. An exact unit contains donor, complete literal source sequence_alignment for both chains, and genotype-aware V/J calls. An amino-acid CDR3 unit, nucleotide fragment, exact receptor pair and author clonal family are different units. All 109 sample-metadata rows have tech_rep=1 and repeated source samples are described as biological. Direct recombinant-antibody labels attach only to tested exact pairs; untested units remain unknown. The frequency score uses a predeclared 0.5-pseudocount logratio, without label training. Same-compartment contrasts retain original day 181 where supplied. No pooled response-confusion matrix or biological FDR is inferred from binding labels.

Table S1. Data sources, units, sample roles and access

Source	Accession / source	Unit and samples	Purpose	Rights and limit
Rytlewski / NoisET gDNA	10.1371/journal.pone.0213684; original provider	Four aliquots, one pool; complete nucleotide fragments; estimated/template-normalized counts	Characterized post-split technical null and benchmark	Research Use Only / no separate permissive redistribution grant verified; acquire from provider
Minervina TCR	PRJNA633317; Zenodo 4065547	Donor M PBMC F1/F2 biological replicates; days 15/37/85; UMI consensus ntCDR3	Longitudinal comparison, published selections, scalability	Retained processed asset CC-BY-4.0; obtain original input
YFV2018 TCR	PRJNA493983; 10.1073/pnas.1809642115	S1 day0 F1→day15 F1; day0 biological blood replicates train; RNA molecules	Independent antigen-reference association	Article CC-BY-NC-ND 4.0; processed-data redistribution rights not separately settled
Kim/Zhou BCR	PRJNA777934; Zenodo 5895181	Donor 368-01a LN1/NS d60 s13/s14, d110 s15/s16; biological replicates; distinct-cell counts, heavy AA+V/J unit	Earlier compatibility; distinct from exact-pair binding	Processed asset CC-BY-4.0
Matz/Zhou BCR	PRJNA1049392; Zenodo 13929847; GSE328165	Donor + full aligned heavy/light variable sequences and V/J; same tissue/sort over time	Selected tested-pair HA-binding association	Processed asset CC-BY-4.0; no family label transfer
YFV2020 annotation screen	PRJNA577794; eLife 53704 source workbook	NS4B-dextramer sequence phenotype; no matched count/time series in acquired workbook	Screened separately; not directional truth	Public CC-BY source; alpha-sheet donor ambiguity retained; not repaired or used as response truth

Acquisition links: Minervina, Kim/Zhou, Matz/Zhou, Rytlewski provider, YFV original study. Rights statements describe retained asset records and do not replace checking current access terms.

Table S2. Representative real TCR posterior summaries

Minervina donor M day15 F1→day85 F1. Counts are UMI consensuses; intervals and location summaries belong to their own models. Practical-equivalence probability `P(|r| <= 0.3)` is defined and emitted only by Extended (`NA` for Core).

Example	Short label (Figure 3)	Exact ntCDR3	Counts	Extended mean	Extended 95% CI	Core median	Core bounds	Extended practical-band probability
Expansion	`TGCAAA…TTTTT`	TGCAAACACCGACTAGCGGGAAACACCGGGGAGCTGTTTTTT	2→13	2.116	0.944 to 2.833	1.512	0.302 to 2.923	0.02443
Zero-compatible	`AAATGC…ATTTT`	AAATGCCCAGTGCTCCTCCTTTACACAGATACGCAGTATTTT	0→1	0.026	0.000 to 0.944	0.000	-0.907 to 1.008	0.9511
Contraction	`TGCAGC…ATTTT`	TGCAGCACGAACGGGGGGCTGGATCAGCCCCAGCATTTT	44→3	-2.899	-3.777 to -1.889	-2.016	-3.024 to -1.109	6.959e-9
Low counts (additional)	`AAAAAA…TTTTT`	AAAAAACTGTTTTTT	1→0	-0.021	-0.944 to 0.000	-0.101	-1.512 to 0.706	0.9530

Table S3. Matched Core-versus-NoisET compatibility benchmark and separated integrated Extended + Core product benchmark

Values are medians of three formal repetitions following one warmup on the evaluated Apple M4 Pro system (12 logical CPUs, 24 GiB RAM, `macOS27.0.1 build26A434`, `arm64`). GB uses 10^9 bytes. Two benchmark scopes are strictly separated: (1) the **matched compatibility benchmark** comparing single-model ClonoShift Core (`clonoshift legacy-response`) against pinned Python NoisET (`noiset_matched.py --stage response`, upstream commit `2b6a45fcd26e7f9eefb3004d81725355cf6d66c9`) from identical input TSVs and identical supplied noise parameters (`model.json`); and (2) the **integrated product benchmark** measuring the full dual-model ClonoShift Extended + Core workflow (own same-study replicate noise fit + Extended + Core + side-by-side comparison tables). Speed ratios are reported only for the matched single-model compatibility benchmark and are computed strictly from the formal medians (`Python NoisET median / ClonoShift Core median`).

Benchmark scope	Tool / workflow	Workload	Union keys	Distinct pairs	Median wall seconds (3-run range)	Median peak RSS bytes (3-run range)	Matched speed ratio (NoisET med / Core med)	Complete output identity & parity
Matched compatibility	ClonoShift Core (`legacy-response`)	gdna	280051	972	1.437000 (1.422000–1.553000)	278000000 (277000000–279000000)	20.171190x (20.2x)	True (PASS)
Matched compatibility	Pinned Python NoisET (`--stage response`)	gdna	280051	972	28.986000 (28.899000–31.466000)	929000000 (923000000–930000000)	1.000000x (reference)	True (PASS)
Matched compatibility	ClonoShift Core (`legacy-response`)	million-real	1055602	3614	5.275763 (5.156978–5.618355)	847626240 (844300288–852312064)	18.634388x (18.6x)	True (PASS)
Matched compatibility	Pinned Python NoisET (`--stage response`)	million-real	1055602	3614	98.310609 (97.935263–99.276722)	2726559744 (2726281216–2735013888)	1.000000x (reference)	True (PASS)
Integrated dual-model product	Full ClonoShift (fit + Extended + Core + comparison)	gdna	280051	972	4.606608 (4.550209–4.794112)	1638645760 (1638645760–1650507776)	Not applicable (dual-model scope)	True
Integrated dual-model product	Full ClonoShift (fit + Extended + Core + comparison)	million-real	1055602	3614	15.840882 (15.543550–16.327939)	5452152832 (5130371072–5545181184)	Not applicable (dual-model scope)	True

The integrated Extended + Core configuration used 33 frequency points, response probability 0.2, scale 1, maximum shift 6, stride 1 and delta 0.3, with certified omission. These supplied settings describe this benchmark, not an empirically calibrated prior. Default worker policy allows up to four workers; the harness does not independently certify a per-run worker count. Initial validation/merged input preparation and post-run external hashing and parity verification are outside the wall timer.

Table S4. Finite Core response compatibility

Same supplied noise; historical Rust executable and pinned Python reference identities are in the private claim ledger. Discrete median/mode/interval differences are zero throughout.

Dataset record	Distinct pairs	Numeric comparisons	Max mean error	Max tail error	Exact selected keys	Status
m13-gdna-python-parity.json	972	9338	2.300e-14	2.225e-13	True	PASS
m13-python-parity-minervina-15-37.json	3507	24548	8.903e-14	8.833e-13	True	PASS
m13-python-parity-minervina-15-85.json	3614	25190	1.429e-13	1.417e-12	True	PASS
m13-python-parity-yellow-0-15.json	1910	14966	1.671e-14	4.996e-15	True	PASS
m13-python-parity-bcr-60-110.json	53	3824	3.553e-15	1.221e-15	True	PASS

Table S5. Same-compartment exact-pair BCR binding association

AUC uses observed-frequency logratio, not a ClonoShift model posterior. NA means no assayed positives. Source repetitions and overlapping exact pairs are not independent truth trials. Shortened Figure 5 labels map directly to the full donor, tissue, sorting-gate, and timepoint identifiers below.

Figure 5 label	Donor	Compartment (tissue/sorting gate)	Time	Labeled pairs	HA positive	HA negative	Binding AUC
397-05 · LN (d0→d181)	397-05	LN/NS	d0→d181	9	1	8	0.812
397-05 · Blood (d0→d180)	397-05	blood/IgDlo	d0→d180	14	0	14	NA
397-06 · LN (d0→d181)	397-06	LN/NS	d0→d181	43	9	34	0.632
397-06 · Blood (d0→d180)	397-06	blood/IgDlo	d0→d180	4	1	3	0.833
397-09 · LN (d0→d121)	397-09	LN/NS	d0→d121	12	1	11	0.636
397-09 · Blood (d0→d121)	397-09	blood/IgDlo	d0→d121	58	0	58	NA
397-17 · LN (d0→d181)	397-17	LN/NS	d0→d181	50	6	44	0.636
397-17 · Blood (d0→d121)	397-17	blood/IgDlo	d0→d121	25	0	25	NA
397-22 · LN (d0→d181)	397-22	LN/NS	d0→d181	43	1	42	0.964
397-22 · Blood (d0→d180)	397-22	blood/IgDlo	d0→d180	86	2	84	0.158
397-28 · LN (d0→d181)	397-28	LN/NS	d0→d181	9	0	9	NA
397-28 · Blood (d0→d180)	397-28	blood/IgDlo	d0→d180	3	0	3	NA
397-29 · LN (d0→d181)	397-29	LN/NS	d0→d181	216	26	190	0.590
397-29 · Blood (d0→d180)	397-29	blood/IgDlo	d0→d180	1	0	1	NA

Table S6. Correlated held-out technical-null comparisons

Conditional NLL; lower is better. Last columns are fractions with model-tail value≤0.05 among totals≥100.

Fold	Training→target	Poisson	Global BB	Bins	Smooth	Bins high-total fraction	Smooth high-total fraction
1	replicate_1_1 + replicate_1_2 → replicate_2_1 + replicate_2_2	0.761783497	0.761426622	0.760466847	0.760497281	0.037736	0.044025
2	replicate_2_1 + replicate_2_2 → replicate_1_1 + replicate_1_2	0.768346566	0.767576133	0.765828214	0.765728425	0.061688	0.032468
3	replicate_1_1 + replicate_2_1 → replicate_1_2 + replicate_2_2	0.767004210	0.766438229	0.764757281	0.764643233	0.015924	0.012739
4	replicate_1_2 + replicate_2_2 → replicate_1_1 + replicate_2_1	0.761076920	0.760283943	0.759539758	0.759519342	0.125806	0.090323
5	replicate_1_1 + replicate_2_2 → replicate_1_2 + replicate_2_1	0.759882458	0.759569927	0.758868489	0.758882859	0.052117	0.032573
6	replicate_1_2 + replicate_2_1 → replicate_1_1 + replicate_2_2	0.768640611	0.767795102	0.766096348	0.765860457	0.056782	0.031546

Table S7. Antigen-reference association and experimental sensitivities

All rows use the 129-node evaluated corrected setting unless labeled Core. Practical positive event is P(r>0.3)>0.95. Core uses its zero-effect positive-tail diagnostic; the two events are not interchangeable.

Method	YFV hits/observed	CMV hits/observed	Event / scope
ClonoShift Extended (hybrid)	5/40	0/12	Practical effect; research association
ClonoShift Core	5/40	0/12	Zero-effect diagnostic; not contraction
Experimental corrected Poisson research projection	6/40	0/12	Practical effect; research association
Experimental NB2 point	0/40	0/12	Practical effect; research association
Supplied Poisson sensitivity	0/40	0/12	Practical effect; research association
Allocation NB2 point	1/40	0/12	Practical effect; research association
Joint bootstrap	1/40	0/12	Practical effect; research association

Table S8. Interpretation and method status

Method	Status	Interpretation boundary
ClonoShift Extended	Supported corrected hybrid numerical inference	Fixed qualified Core parameters; supplied prior; no empirical biological FDR
ClonoShift Core	Supported bounded compatibility inference	Pinned Poisson formulation; bounded endpoint checks; zero-inclusive tail
Model comparison	Supported descriptive comparison	Mean/median signs; no consensus biological verdict
Automatic corrected fitting / NB2	Experimental	Real automatic fits refused; selected NB2 predictions improved
Joint fitted-parameter uncertainty	Experimental	Observed-pair sensitivity; no nominal coverage theorem
Adaptive parameter grid	Experimental unresolved	Modes/tails not resolved as a continuous posterior
Abundance-aware conditional null	Experimental	One gDNA pool; high-total tail failures; unknown true exposures
BH/BY and empirical ranks	Diagnostic	Valid input null values and assumptions remain necessary
TCR antigen / exact-pair BCR binding	Experimental association	No exhaustive true directional labels; no family phenotype propagation
Calibrated biological or BCR-family response	Unsupported / refused	Assay-matched design and independent truth are absent

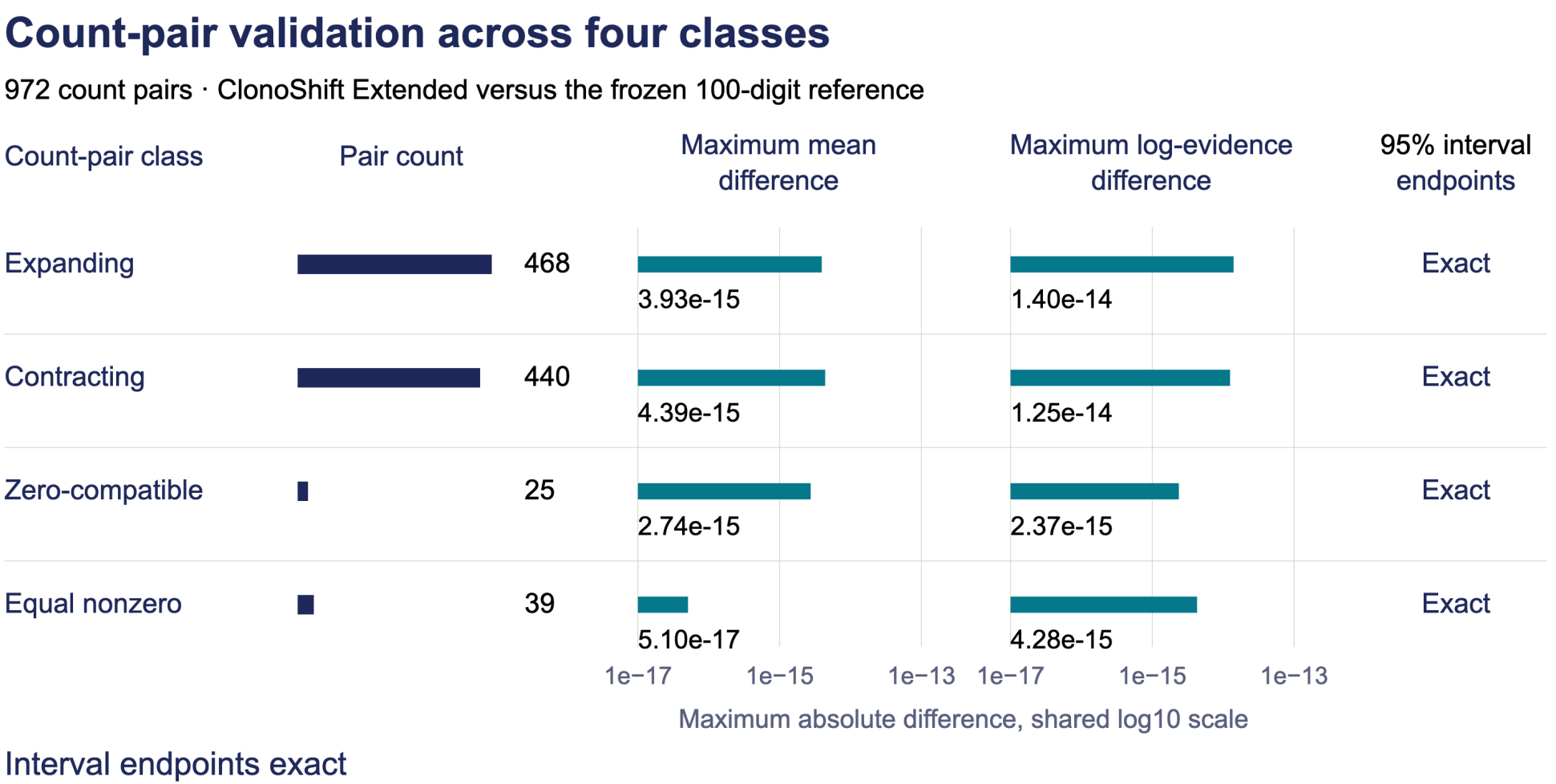


Figure S1. Extended numerical validation across four count-pair classes. A compact validation matrix shows expanding (468 pairs), contracting (440 pairs), zero-compatible (25 pairs), and equal nonzero (39 pairs) classes from the same independent 100-digit decimal oracle used in Figure 2a. Pair-count mini-bars are proportional on a common linear scale; thin horizontal bars show maximum posterior-mean and log-evidence absolute differences on a shared log10 scale. Direct numerical labels retain each maximum and the status column reports exact credible interval endpoints for every class. All 972/972 count pairs were validated and all 972/972 credible interval endpoint pairs matched exactly. These are numerical reference checks, not independent biological calibration.